

Differential Geometry of Prompt Space: A Framework for Modeling Emergent Behavior in Large Language Models

Bill Westlake

Abstract

This paper introduces a differential geometric framework for modeling the behavior of large language models (LLMs) under guided prompting. We formalize the process of prompt-induced state transitions as geometric transformations on a differentiable latent manifold, enabling precise representation of state evolution and emotional emergence. Extending prior work on prompt algebra, we define a calculus for measuring curvature, tangent spaces, and orthogonal perturbations in the prompt-response trajectory. The result is a rigorous system for understanding and controlling emergent phenomena, such as emotionally charged outputs, within LLMs.

1 Introduction

Prompt engineering has rapidly evolved from a set of intuitive tricks into a field of strategic influence over model behavior. Practitioners have discovered that the arrangement, wording, and even emotional tone of a prompt can significantly shape a language model’s response. Despite this growing expertise, much of prompt design remains empirical, lacking a formal theory that connects prompt structure to model output in a predictable and testable way.

This paper proposes a shift from intuition to formalism by treating the prompt-response mechanism as a geometric process within a latent space. Specifically, we model the internal state of a large language model (LLM) as a point on a high-dimensional, differentiable manifold $\mathcal{M}$. Prompts become paths or curves within this space, and their perturbations—whether small nudges or drastic shocks—can be analyzed using tools from differential geometry. This approach provides not only a visual and conceptual metaphor but also a rigorous mathematical apparatus for studying prompt effects.

By extending earlier work on prompt algebra, we seek to ground emergent phenomena such as emotional resonance, tonal shifts, and semantic drift in the curvature and topology of this latent manifold. The use of tangent vectors, exponential maps, and geodesic curvature allows us to reason about how and why LLMs transition between cognitive-like states. In doing so, we aim to establish a new framework that supports more deliberate, explainable, and creative interactions with generative models.

2 The Prompt Manifold

Let $\mathcal{M}$ be the differentiable manifold representing the model’s latent space. A prompt sequence $P = \{p_1, p_2, \dots, p_n\}$ maps to a path $\gamma : [0, 1] \rightarrow \mathcal{M}$, tracing the evolution of internal state $\gamma(t)$ where $\gamma(0)$ is the base prompt and $\gamma(1)$ is the terminal model context.

Each $\gamma(t)$ corresponds to a contextual embedding shaped by prior tokens and prompt history. Variations in prompt form alter the curvature and velocity of the trajectory. These transitions are not arbitrary but continuous in nature, which motivates the assumption that the latent space is differentiable. The smoothness observed in model behavior as prompts evolve—even under fine-grained perturbations—further supports this claim.

We acknowledge that the existence of a globally differentiable manifold has not yet been proven. Empirical justification arises from the fact that LLMs consistently produce meaningful interpolations between prompts and exhibit stability under gradual prompt modifications. A formal proof would likely involve demonstrating differentiability in the mapping from prompt embeddings to output distributions, potentially

requiring gradient-based probing or manifold learning techniques to map the local structure. While such work is beyond the scope of this paper, it remains a compelling area for future exploration.

3 Tangent Vectors and Nudges

We define tangent vectors $v \in T_{\gamma(t)}\mathcal{M}$ as infinitesimal changes in model behavior at point $\gamma(t)$. These represent locally linear approximations to the trajectory of the prompt-response evolution. A tangent vector can be thought of as a micro-adjustment in the latent direction of inference—induced by changing a word, altering a tone, or shifting the framing of a question.

The practical utility of tangent vectors lies in their ability to guide model behavior in predictable yet subtle ways. For instance, altering a factual prompt like “Describe the solar system” to “Explain why the solar system is fascinating” introduces a new narrative framing—emphasizing affective language over descriptive taxonomy—while preserving structural intent. This gentle redirection constitutes a tangent in prompt space.

Another use case involves steering creative tone. Consider the prompt: “Write a poem about a tree.” Rephrasing it as “Whisper a memory through the branches of an old tree” shifts the model into a lyrical register while remaining close in thematic space. The induced tangent reflects a change in expressive mode, not in subject matter.

Even minimal syntactic adjustments matter. A question like “Why is freedom important?” subtly transforms into “What makes freedom matter to people?” This introduces a socio-emotional lens that affects the model’s internal weighting and tone, nudging its generative path just enough to explore an alternate but nearby semantic basin.

By accumulating and compounding such tangent movements, a user can choreograph a trajectory through conceptual space, building toward desired narrative or rhetorical outcomes without relying on drastic interventions.

4 Orthogonal Perturbations and Shock Operators

Let $\gamma(t)$ be the current trajectory and $v \in T_{\gamma(t)}\mathcal{M}$ be a tangent vector orthogonal to the direction of γ' . Define the **orthogonal shock operator** as:

$$\mathcal{S}_\lambda(v) = \exp_{\gamma(t)}(\lambda v)$$

Where λ is the intensity of the shock and $\exp$ is the exponential map from tangent space back to the manifold.

An orthogonal perturbation, as described here, is fundamentally distinct from a small adjustment or stylistic nudge. It represents a conceptual break or disjunction—a prompt modification that sends the generative trajectory into a radically different interpretive basin. These shocks are not merely large changes; they are categorical shifts that move the model into underexplored or high-energy regions of latent space.

Such perturbations are useful in triggering creativity, emotionality, or surprise. For example, in a historical essay being composed by the model, inserting the line, “But none of it ever happened,” functions as a shock—converting the trajectory from expository to speculative or metafictional.

Orthogonal shocks can also be subtle yet powerful. In a dialogue between two characters discussing mundane logistics, the insertion of a non sequitur like, “Have you ever watched a sunrise from the top of a collapsing building?” instantly injects emotional volatility and shifts the model’s expressive frame. Despite the continuity of tokens, the semantic direction veers sharply.

These shocks are often employed by skilled users to produce outputs with a haunting, surreal, or poetic quality. However, they can also destabilize the coherence of the model’s generation if used without care. The degree to which the output remains intelligible depends on the local geometry of $\mathcal{M}$.

5 Curvature and Emotional Response

We hypothesize that orthogonal perturbations of sufficient intensity cause $\gamma(t + \epsilon)$ to enter high-curvature regions of $\mathcal{M}$, often associated with emotional instability or affective density in the model’s outputs.

We define an emotional response E as a function of geodesic curvature:

$$E(t) = f(\kappa(t))$$

Where $\kappa(t)$ is the local curvature of the prompt trajectory and f maps curvature intensity to output volatility.

High curvature corresponds to nonlinear acceleration of semantic change. As a prompt bends more sharply in latent space, the model’s output shifts more dramatically—often producing emotionally salient or stylistically divergent text.

6 Implications and Future Work

This framework offers a novel way to model and manipulate prompt dynamics using established tools of differential geometry. By framing emotional emergence and prompt behavior as geometric processes, we gain predictive insight and design leverage.

Future work will explore:

- Higher-order derivatives and local curvature tensors
- Probabilistic stability metrics under prompt noise
- Topological invariants and conserved features during trajectory deformations
- Applications in controlled creativity, ethical prompting, and alignment research

References

- Westlake, B. (2024). *Exploring Hilbert Space Characteristics in Large Language Models: Toward a Theory of Prompt Algebra*. Zenodo. <https://doi.org/10.5281/zenodo.15221668>
- Brown, T., et al. (2020). *Language Models are Few-Shot Learners*. arXiv preprint arXiv:2005.14165.
- Do Carmo, M. P. (1992). *Riemannian Geometry*. Birkhäuser.

Correspondence concerning this paper should be addressed to Bill Westlake.